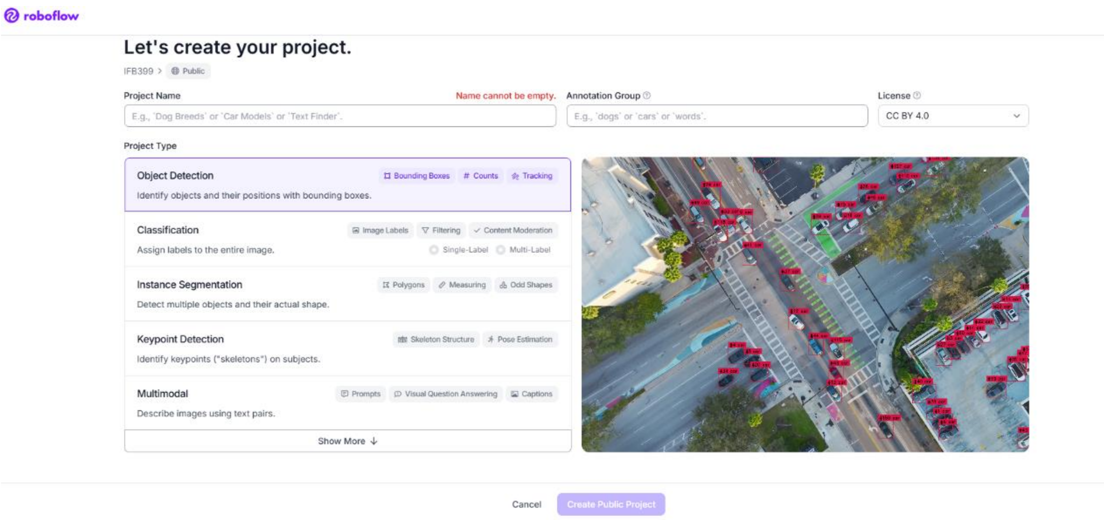
Instructions for Training and Exporting Datasets

The program that we have used to train our models is called Roboflow. Roboflow is an end-to-end computer vision platform designed to help developers and organizations build, train, and deploy custom vision models with ease. It streamlines the entire machine learning workflow—from data collection and annotation to model training and deployment—without requiring deep expertise in AI or computer vision. The free version of Roboflow only contains 30 credits per month. If you want to have more than the amount allocated, it is recommended to purchase one of their subscriptions.

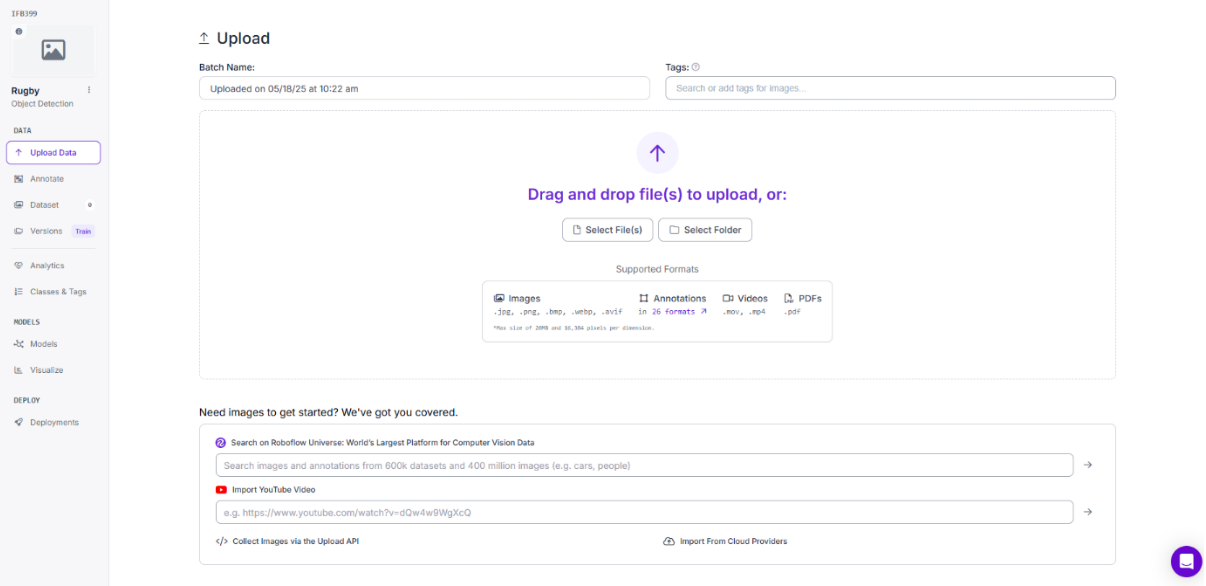
To generate the ruck, lineout and ball models, we annotated and created our datasets in Roboflow, and then using Google Colab we trained the models. The following instructions outline how you can update and expand datasets by adding more images or create a new dataset.

Step 1. Create a New Project

To begin, google Roboflow, press 'Get Started' and sign up with a free account. To start training your dataset, you need to create a new project, give it a name and select *Object Detection* as the project type.

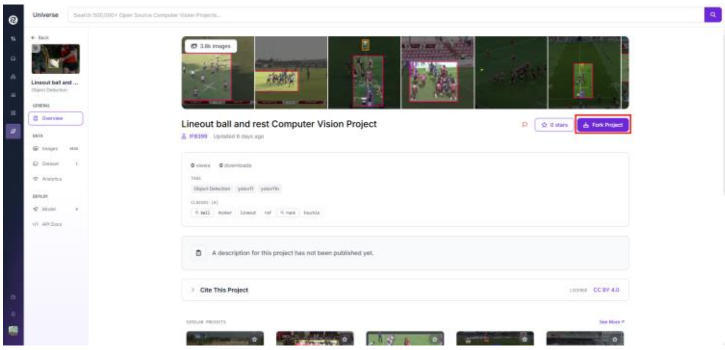


This will take you to the main project page:

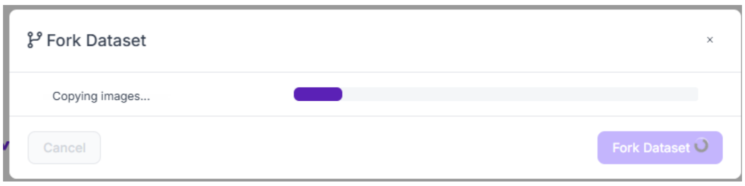


Step 2: Getting previous dataset to your Roboflow

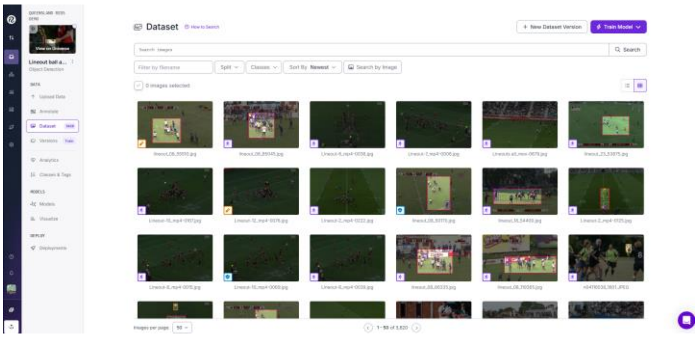
The handover artefact includes a document which contains links to our previously created models in Roboflow. To get these datasets, follow one of these links and select *Overview* in the menu tab and then click on *Fork Project* (you must be signed into Roboflow).



Select *Continue* and it will start to download that dataset into a new project.



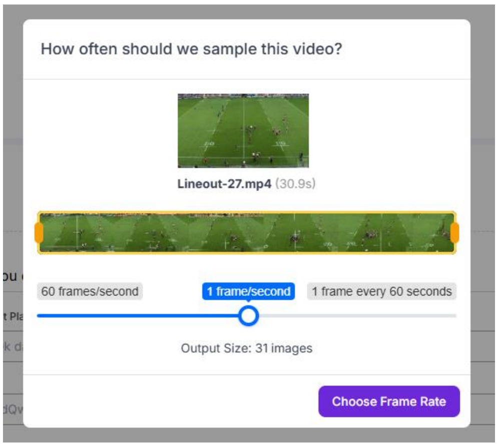
You should now have your own version of that model.



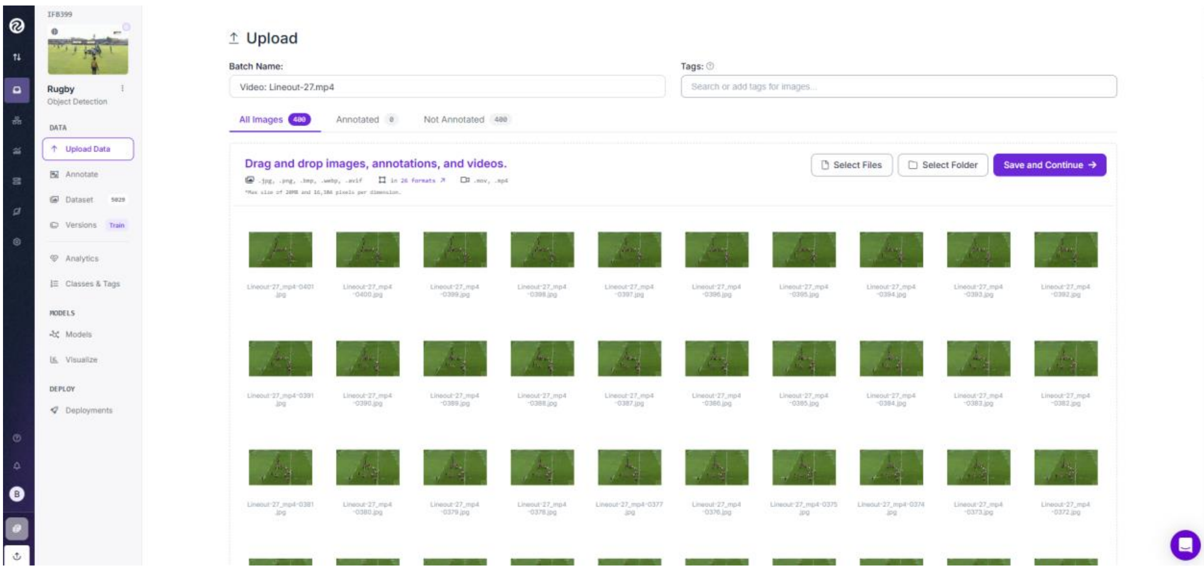
Step 3. Uploading Images for annotation

Once you have set up the project, you can add additional images to the dataset and continue to train the model to make it more accurate. To start off, click on *Upload Data* to upload a video or a collection of images you want to include.

If you are including a video, select the frame rate at which you want the video to be sampled at. This will extract a number of photos from the video.



Once this is processed, you can go through and delete any of the created images that you don't want to include in the dataset.



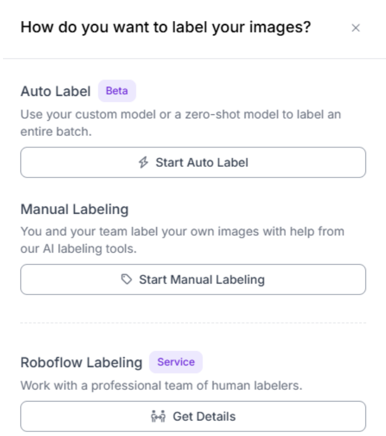
Then click *Save and Continue*. This will upload the images properly which will then allow you to annotate them.

Step 4: Annotating Images

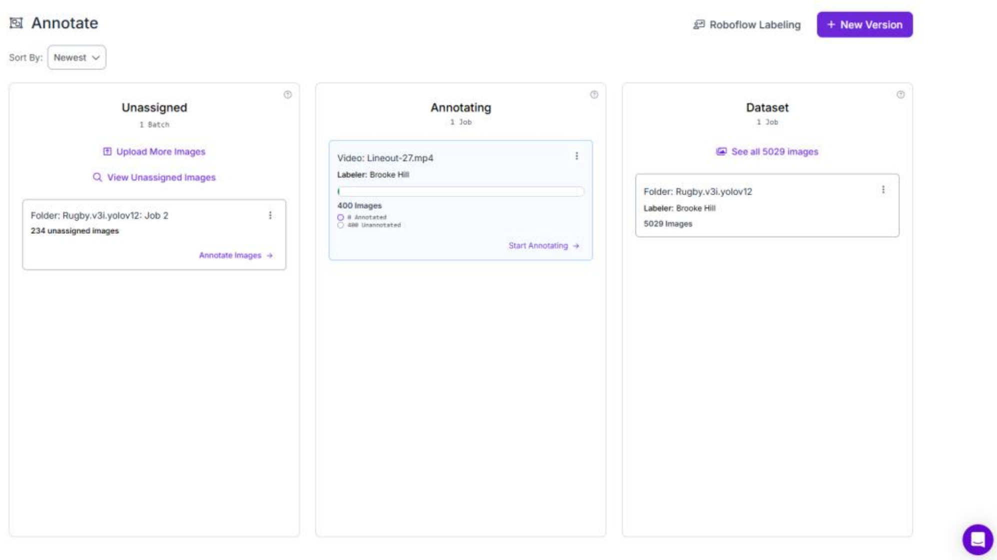
There are a couple of ways you can annotate images - manually or by using a Roboflow model to assist with annotations.

4.1 Manual Annotations

When prompted with a "How do you want to label your images?" question, select *Manual Labelling*.



Select *Annotating* in the centre column:



Click on the first image and an annotations window will appear.

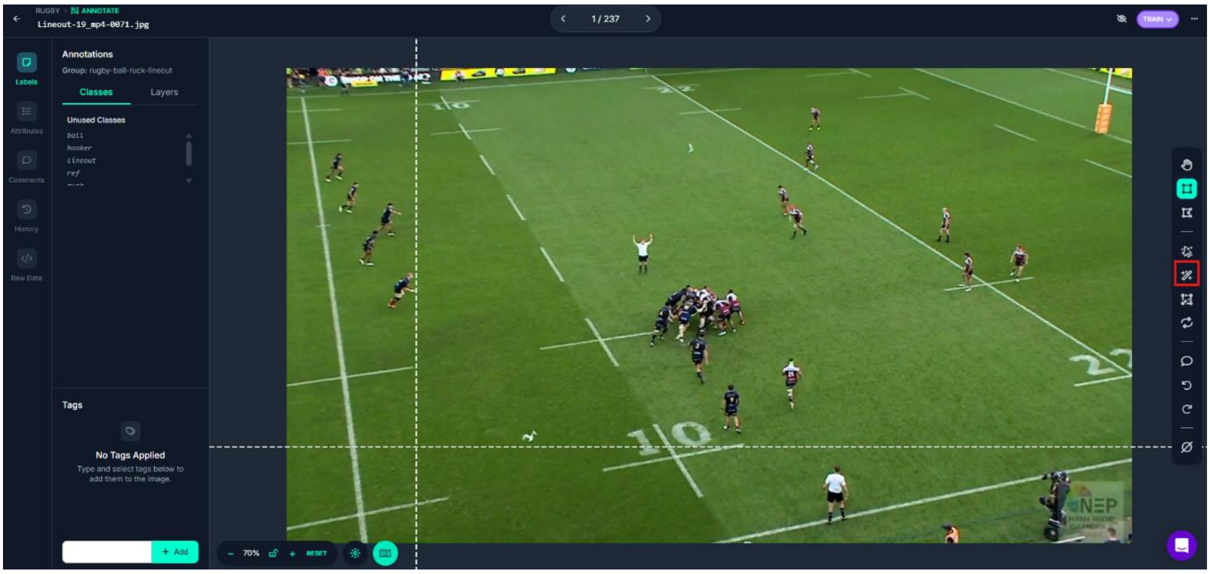
Draw a box around the point you want to annotate. In this instance it is the referee:

Repeat these manual annotations for the remainder of the images, dragging and drawing boxes around the referee. You can add multiple annotations to the one image, when appropriate.

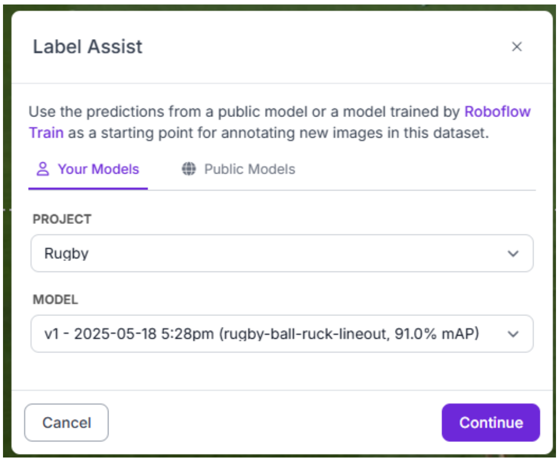
4.2 Using Pre-Trained Model to Assist with Annotations

To make it easier, you can opt to use pre-trained models that are based on older versions of the dataset to help annotate images, making the process faster and less tedious. Please ensure you have trained a model previously in Roboflow – see step 7.

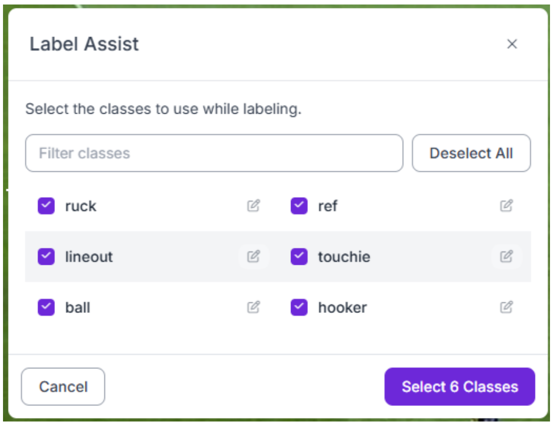
Whilst on the annotations screen, navigate to the right-hand side and select the *Label Assist* (wand icon) tool (highlighted in red below).



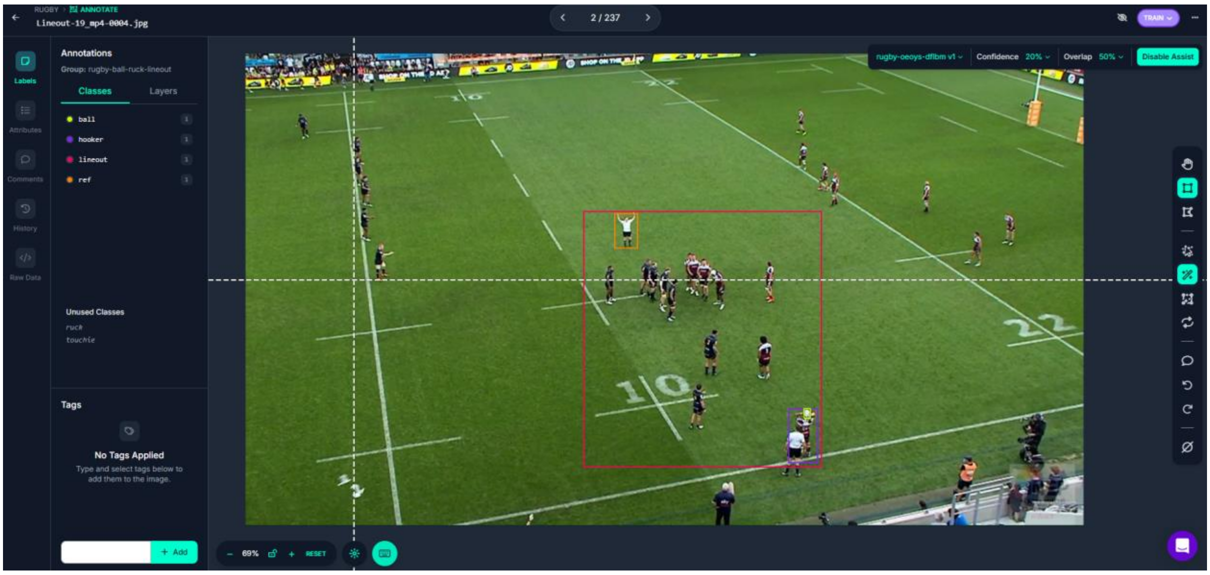
Select your most recently trained model.



Select the classes you wish to assist in the annotations.



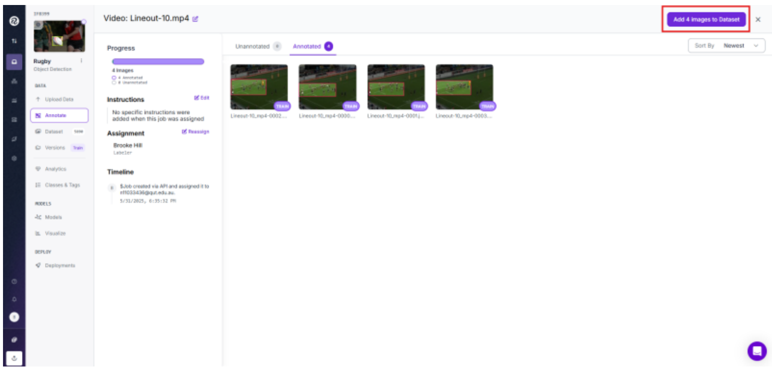
If the model is trained to detect the image classes you have selected, it will automatically annotate each image. You can view these annotations by clicking on each image individually.



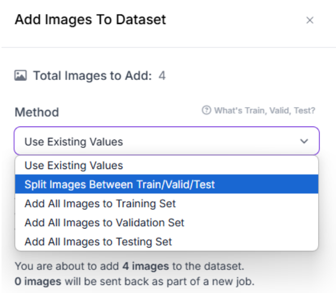
Step 5: Add Annotated Images to Dataset

Once you are satisfied with the annotations, you can add them to an existing dataset. You will need to retrain the model for it to include the new annotated images.

To add images into the dataset, navigate to the *Annotated* images tab and in top right corner select *Add X Images to Dataset*.



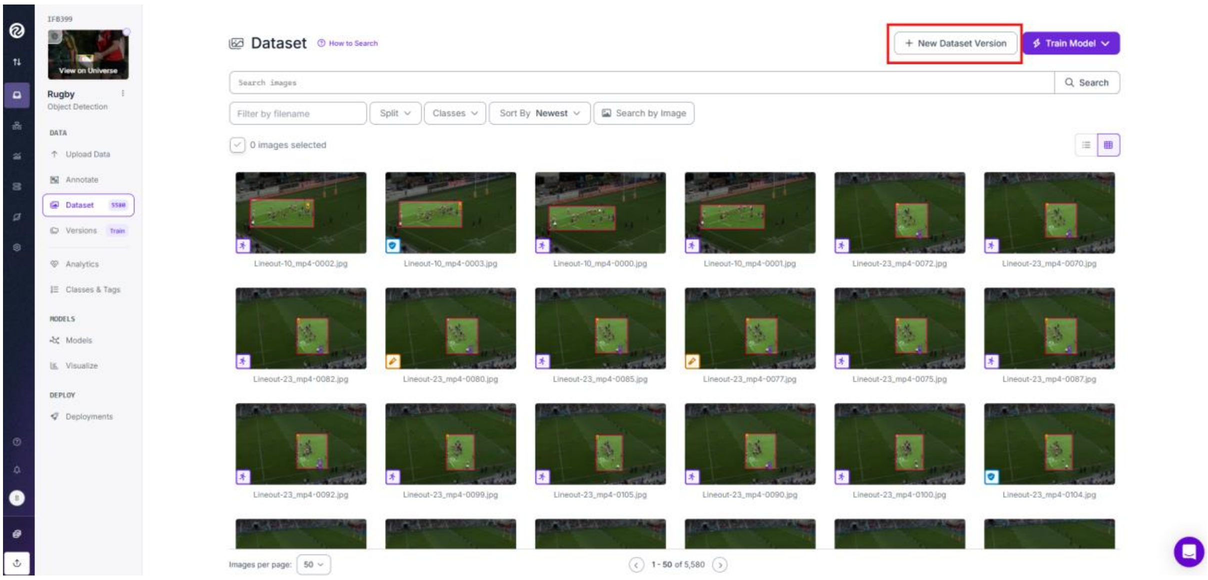
Next, you want to select where you want those images to be added to. Generally, we chose to *Split Images Between Train/Valid/Test* option.



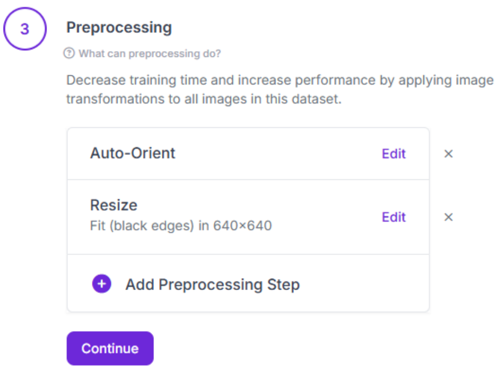
After you have selected the location to add the images to, press *Add Images* – this will move the images into your dataset.

Step 6: Creating Dataset Version

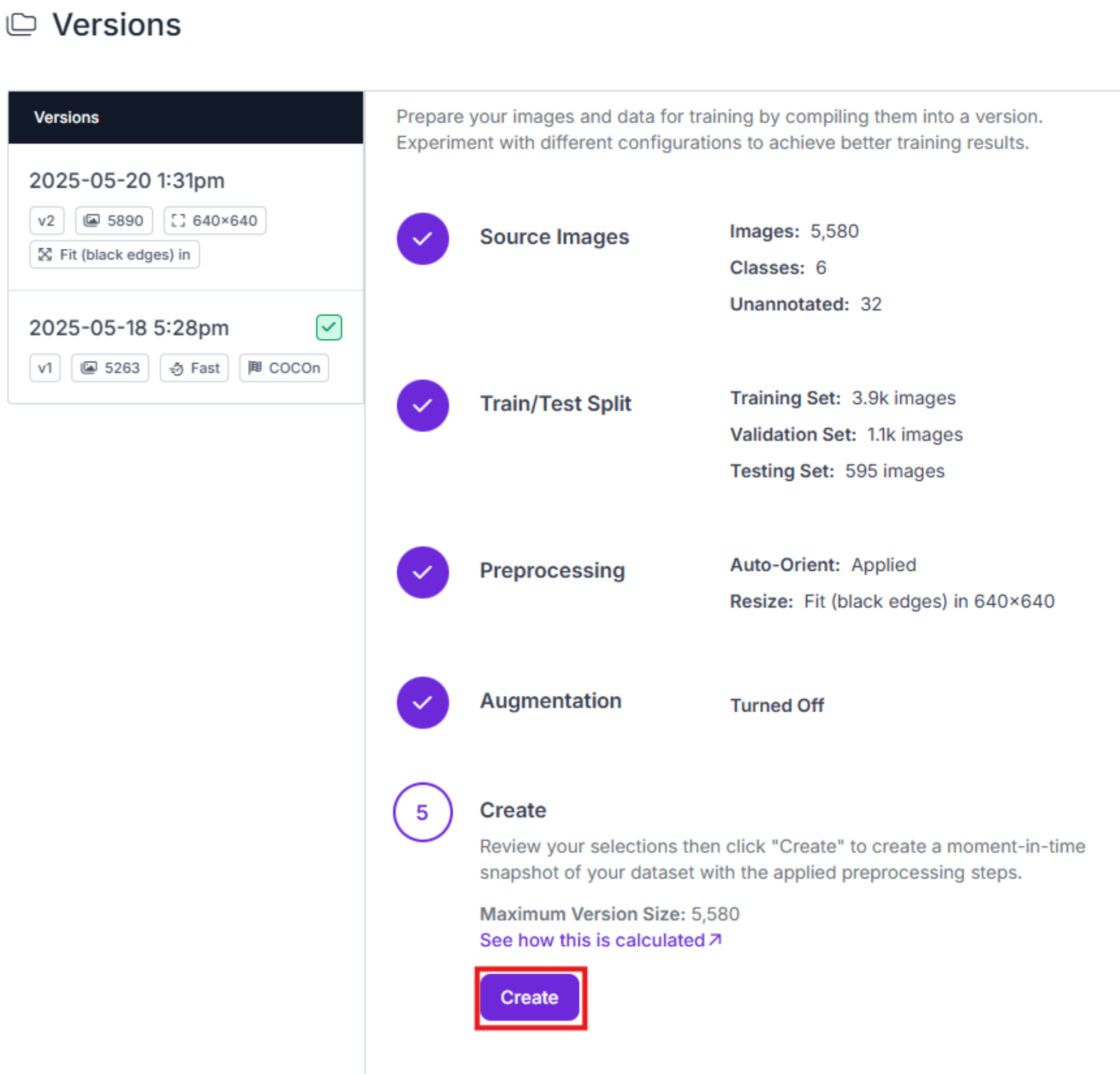
After you are happy with your dataset, you can create a *New Dataset Version* to use for model training.



On the next page, select the features you want for your version. For this, we used the standard 640x640 size for all images (fit within this size with black edges). Make sure not to select the *Reflect Edges* mode, as it produced undesirable outputs.



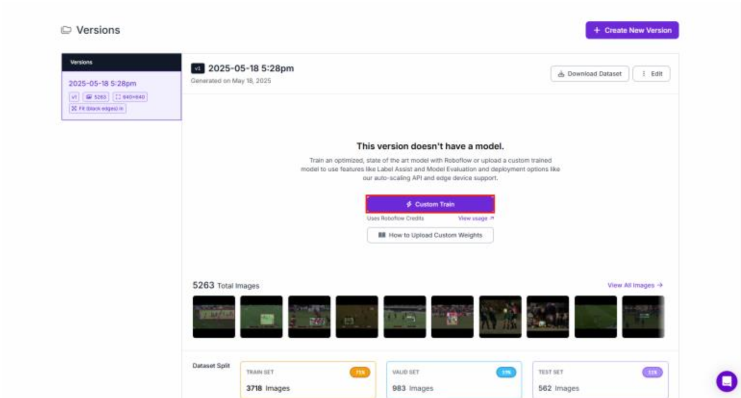
After selecting all wanted preprocessing, select *Create*:



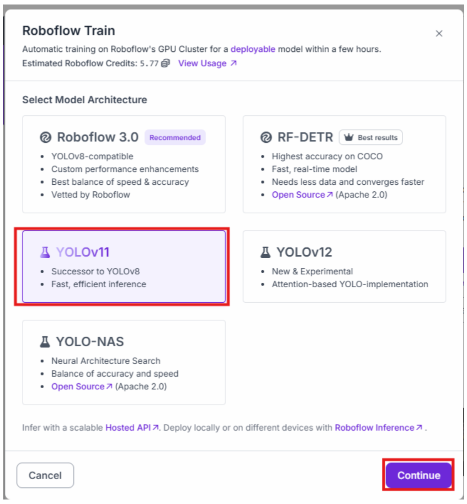
You have now created a dataset version. Hooray!

Step 7: Train the Model in Roboflow – (assistance on annotating images)

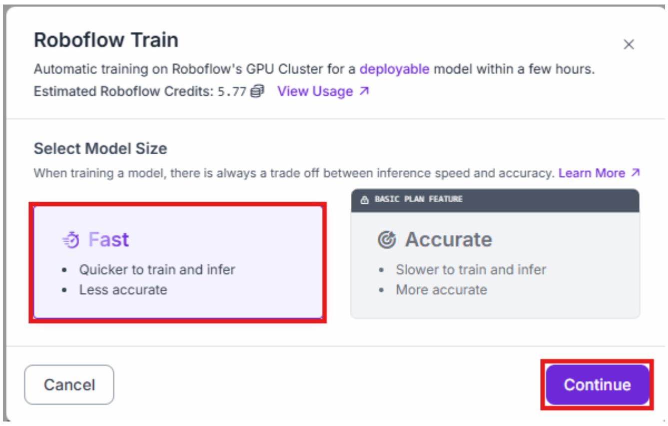
Using your dataset, click on *Custom Train* to begin the process of training the model in Roboflow.



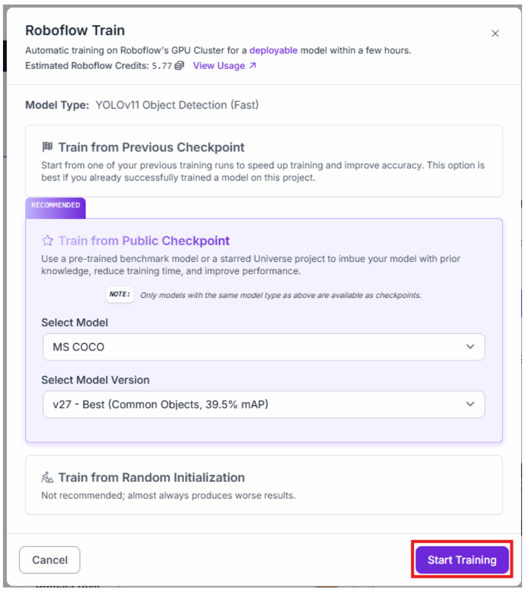
It is recommended to use the YOLOv11 model, and then select *Continue*.



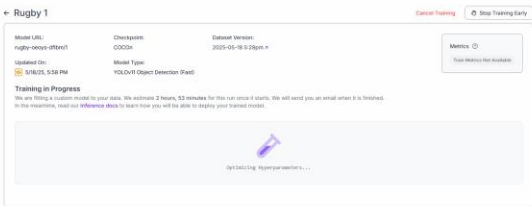
Choose the *Fast* option.



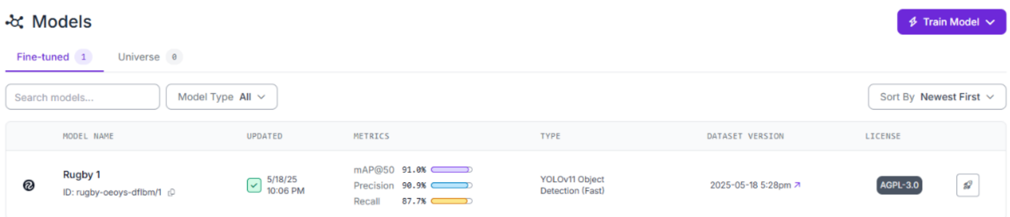
Then click *Start Training*.



The model will begin training.



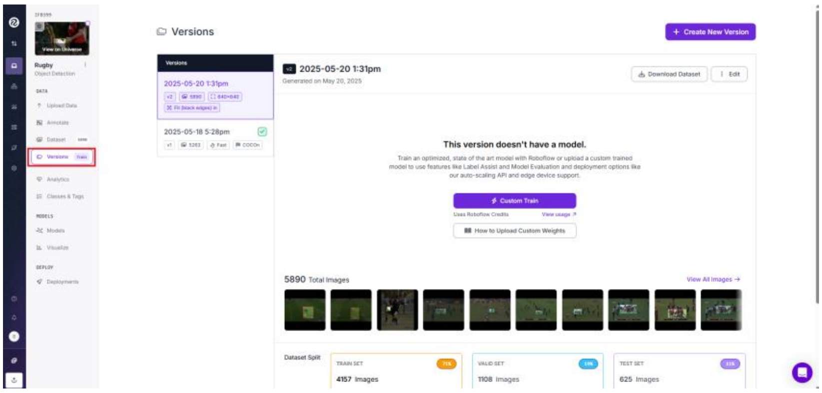
After the model has finished training, you can use it to assist with annotating images. You can now follow the steps outlined in [step 4](#) on how to use the model to assist with annotating extra images for the dataset.



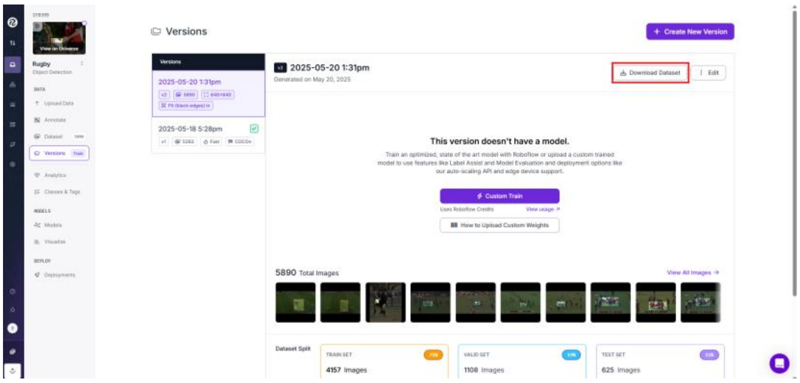
Step 7.1: Getting Code for Model Training in External Coding Environments

To acquire an executable code summary which allows you to train the dataset externally, first ensure that you have created a dataset version, following the steps outlined in step 6.

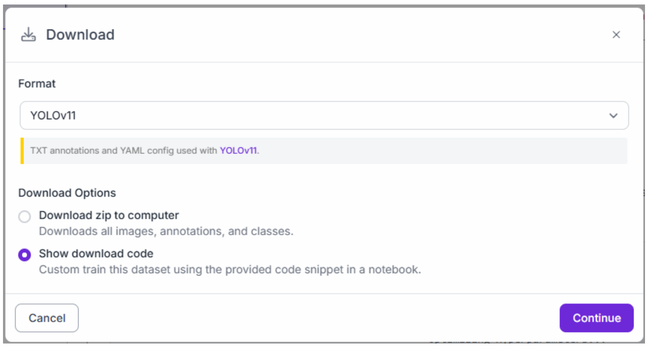
Once this step is complete, click on the *Versions* tab located on the left-hand side menu:



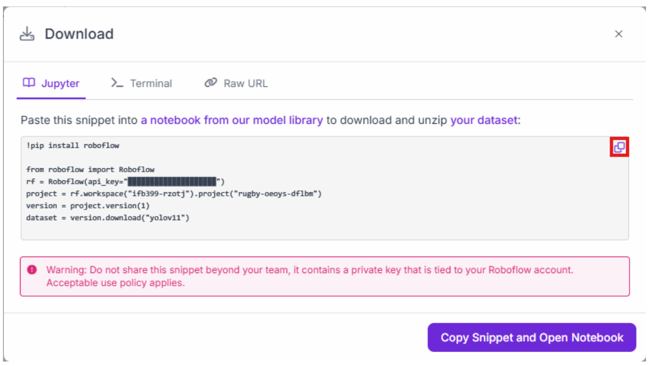
Next, select *Download Dataset*.



After this, select the downloaded code format and click *Show Download Code*. Select *Continue*.



Select the copy button on the top right-hand side of where the code is and move on to step 8.



Step 8: Downloading dataset and training model in Google Colab

While Roboflow does allow for model training, this uses a large amount of the allotted monthly credits. To avoid this, Google Colab offers an online coding environment with a virtual GPU which can train these models (this process can be done in any coding environment, however, if there is no GPU, it can take several hours to complete this step).

Open Google Colab and sign-in with a Google account. Once signed-in, select the File tab, then ‘upload notebook’ and choose the provided training_template.ipynb file. At the top, select the Runtime tab, then ‘change runtime type’. Select Python 3 in the runtime type drop-down menu, then select the T4 GPU as the hardware accelerator and save.

Scroll down to the empty cell (this should be after !pip install roboflow) and paste the copied code snippet from Step 7.1. In the last line, add a comma after “yolov11” and type a dataset name in quotation marks. For example:

```
[ ]  !pip install roboflow

     from roboflow import Roboflow
     rf = Roboflow(api_key="VGLI7GAsSYzFV4E3Ucw1")
     project = rf.workspace("ifb399-rzotj").project("ball-1dnbd")
     version = project.version(1)
     dataset = version.download("yolov11", "ball_dataset")
```

In the final cell (should start with !yolo train), adjust the data file path by adding the dataset name you entered above after content/. For example:

```
!yolo train task=detect model=yolo11n.pt data="/content/ball_dataset/data.yaml" epochs=30 imgsz=640
```

Once this has been updated, select the runtime tab at the top of the page and select ‘run all’. The final cell will usually take 20-30 minutes to run. It is recommended to keep the page open while it is running, as the free version of Colab does not allow background processing.

To obtain the best results, you want the training to run until convergence. This means it will run until the performance plateaus. The easiest way to monitor this is by checking the mAP50 value in each epoch. If it remains the same after many epochs, it has likely converged,

Once training is complete, if the model has not converged, it may be worth training for a longer period to obtain better performance. To run training for longer, increase the number of epochs in the yolo train line and run the final cell again by selecting the play button to the left.